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SECTION B

8. When performing redox titrations the concentration of one solution is found by reaction with another solution of known concentration. Some oxidising agents are classed as primary standards as their concentrations can be calculated precisely from the mass used. Other oxidising agents need to be standardised by reaction with solutions of known concentration before use.

(a) Potassium dichromate, $K_2Cr_2O_7$, is a primary standard. Calculate the concentration of a solution of potassium dichromate made from 5.00 g of potassium dichromate in 250 cm^3 of deionised water. [2]

concentration = ... $mol\,dm^{-3}$

(b) Sodium thiosulfate, $Na_2S_2O_3$, can be standardised using potassium dichromate. To do this, acidified potassium dichromate is used to oxidise iodide ions to iodine molecules and the sodium thiosulfate is then used to reduce the iodine formed.

$$I_2 + 2e^- \rightleftharpoons 2I^-$$

$$2S_2O_3^{2-} \rightleftharpoons S_4O_6^{2-} + 2e^-$$

(i) Write the half-equation for acidified dichromate ions acting as an oxidising agent. [1]

....

(ii) Write an equation for the oxidation of iodide ions to iodine molecules using acidified dichromate. [1]

....

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(iii) Excess KI(aq) was added to 25.0 cm^3 of the aqueous potassium dichromate from part (a) with a small amount of dilute acid. This produced iodine. Aqueous sodium thiosulfate was added from a burette and 24.30 cm^3 of $Na_2S_2O_3$(aq) was needed for complete reaction with the iodine.

I. The titration used an indicator. Name the indicator used. [1]

....

II. Calculate the concentration of the aqueous sodium thiosulfate. [3]

concentration = ... mol dm^{-3}

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(c) Addition of aqueous sodium hydroxide to the solution of potassium dichromate causes a colour change.
State the colour change and explain why this is not a redox reaction. [2]

....

....

....

(d) Heating solid potassium dichromate can cause it to decompose releasing oxygen gas.

$$4K_2Cr_2O_7 \longrightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$$

$$\Delta H^\theta = 348\ \text{kJ mol}^{-1} \quad \Delta S^\theta = 385\ \text{J K}^{-1}\ \text{mol}^{-1}$$

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta$/kJ mol^{-1}
K_2CrO_4	−1382
Cr_2O_3	−1128

(i) Calculate the standard enthalpy change of formation of potassium dichromate, $K_2Cr_2O_7$. [2]

$\Delta_f H^\theta$ = ... kJ mol^{-1}

(ii) Calculate the minimum temperature required for this reaction. [2]

minimum temperature = ... K

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